

COMP 3608 Project Write Up

Predicting Valorant Champions Tour Match Outcomes

A Comparative Machine Learning Approach to Esports Match Prediction

Project Type: B-Rank Mission | Binary Classification | Kaggle VCT Data Subsets

Models Compared: Logistic Regression, Random Forest, XGBoost, with Majority Baseline

Problem Definition

This project focuses on predicting match outcomes in the Valorant Champions Tour (VCT). VCT is Riot Games' professional Valorant esports circuit, where teams compete across regional leagues and international tournaments. Match outcomes affect standings, tournament progression, preparation decisions, and team reputation.

The main problem is: given two competing VCT teams and their historical information before a match, can a machine learning model predict whether Team A will win or lose?

This is a useful problem because esports teams and analysts often need to understand not only which team is more likely to win, but also which factors appear to influence that prediction. The project therefore aims to build and compare models that can predict match outcome while also allowing the group to explain model strengths and weaknesses.

Stakeholders and Needs

Stakeholder	Need
Coaches and analysts	Identify likely match difficulty and support pre-match preparation.
Team managers	Use performance trends to support roster, scouting, and planning decisions.
Tournament organizers	Understand expected match competitiveness for scheduling and event planning.
Commentators and media teams	Use win probability and key factors to support pre-match discussion.

Problem Classification

The project is formulated as a binary classification task because the outcome of interest has two possible classes:

- 1 = Team A wins the match.
- 0 = Team A loses the match, meaning Team B wins.

Predicted probabilities are also used as an interpretive output. For example, a model may predict that Team A has a 0.63 probability of winning, even though the final predicted class is still win or loss.

Data and Experimental Design

The project uses Kaggle Valorant Champions Tour data. To satisfy the B-Rank structure and keep the scope manageable, the work uses three chronological yearly data subsets from the VCT dataset: 2023, 2024, and 2025. These are treated as D1, D2, and D3 in the experimental design.

Data Subset	Season	Matches	Purpose
D1	2023	330	Early historical data used for training.
D2	2024	434	Additional historical data used for training and validation.
D3	2025	501	Future-season holdout test set.

The cleaned master dataset contains 1265 matches and 47 columns. The class distribution is close to balanced, with Team A winning 664 matches (52.5%) and Team B winning 601 matches (47.5%). This means the models are not dealing with a severe class imbalance.

Experimental Design

A chronological split was used rather than a random split. This is important because sports and esports data are time dependent. In a real prediction setting, a model would be trained on past matches and used to predict future matches. Random splitting could leak future information into the training process.

- Main training set: 2023 and 2024 matches.
- Main testing set: 2025 matches.
- Walk-forward validation: train on 2023 and test on 2024, then train on 2023-2024 and test on 2025.

Proposed Solution and Feature Engineering

The proposed solution is to build a match outcome prediction pipeline that converts raw match information into pre-match team strength and context features. The models then use these features to predict whether Team A wins.

Final Feature Direction

The notebook engineers 13 pre-match features. The most important design choice is that historical features are calculated using only matches that happened before the match being predicted. This avoids data leakage.

Feature Group	Examples	Purpose
Historical strength	Team historical win rate, historical average rating	Measures how strong each team has been before the match.
Differential features	hist_win_rate_diff, hist_rating_diff, map_win_pct_diff	Directly compares Team A and Team B.
Map performance	Historical map win percentage	Captures map-level consistency and comfort.
Match context	stage_stakes, elimination match, grand final flag	Captures pressure and tournament context.
Draft context	ta_ban_first	Tests whether draft order adds useful signal.

Differential features were especially important because match prediction is a relative problem. A team can have a good win rate, but what matters most is whether that team is stronger than the opponent in the same match.

Mathematical and Algorithmic Approach

Optimization Framing

Each match is represented by a feature vector x . The target variable y is either 1 or 0, where 1 means Team A wins and 0 means Team A loses. The goal is to learn a function $f(x)$ that estimates $P(y = 1 | x)$, the probability that Team A wins.

The classification objective is to minimize prediction error, using binary cross-entropy/log loss for probability-based learning. In simple terms, the models are penalized more when they make confident but wrong predictions.

Algorithms Selected

Model	Reason for Inclusion	Main Strength	Main Limitation
Majority Baseline	Provides a minimum comparison point.	Shows what happens without learning.	No real predictive intelligence.
Logistic Regression	Interpretable baseline for binary classification.	Easy to explain through coefficients.	May miss nonlinear patterns.
Random Forest	Nonlinear ensemble model.	Can capture interactions and gives feature importance.	Can overfit and may be less stable across seasons.
XGBoost	Advanced boosting model for tabular data.	Often strong predictive performance.	Less interpretable and needs tuning.

The three required intelligent algorithms are Logistic Regression, Random Forest, and XGBoost. The Majority Baseline is not counted as one of the three main models; it is used only for comparison.

Implementation Summary

The project was implemented as a reproducible pipeline. The raw VCT files were organized by season, then preprocessing scripts were used to aggregate, clean, and transform the data into a modelling-ready master dataset. The notebook then loaded the cleaned dataset, engineered features, trained models, and evaluated results.

Implementation Pipeline

1. Download and organize the selected 2023, 2024, and 2025 VCT data.
2. Aggregate match-level files into combined category files.
3. Create the master dataset by merging relevant match, team, and performance columns.
4. Clean the master file, handle missing values, and create the binary target variable.
5. Engineer pre-match historical and context features.
6. Train the baseline and three machine learning models.
7. Evaluate models using forward testing and compare model tradeoffs.

Reproducibility

The project is hosted in a GitHub repository with preprocessing scripts, a modelling notebook, and a requirements file. The notebook is intended to be the detailed technical record, while this write up summarizes the project in a format aligned with the assessment rubric.

Results

The models were evaluated using accuracy, F1 score, and ROC-AUC. Accuracy measures overall correctness. F1 balances precision and recall. ROC-AUC measures how well the model separates likely winners from likely losers across different thresholds.

Walk-Forward Results

Fold	Training Data	Test Data	Best Model by F1	Observation
1	2023	2024	XGBoost	Performance was modest because the model had only one season of training data.
2	2023 and 2024	2025	XGBoost	Performance improved after adding more training data.

Final 2025 Test Results

Model	Accuracy	F1 Score	ROC-AUC	Interpretation
Majority Baseline	N/A	0.685	0.500	High F1 is misleading because it predicts Team A every time.
Logistic Regression	0.5988	0.6516	0.6596	Best ROC-AUC and most interpretable.
Random Forest	0.5968	0.6100	0.6263	Useful nonlinear model, but weaker final results.
XGBoost	0.6088	0.6680	0.6270	Best F1 score but less interpretable.

The Majority Baseline has a high F1 score because it predicts every match as a Team A win. This inflates recall, but its ROC-AUC of 0.500 shows that it has no real ability to separate wins from losses. For this reason, ROC-AUC is important when interpreting the results.

Discussion and Tradeoffs

The results show that historical pre-match data can predict VCT outcomes better than random guessing, but the task remains difficult. Valorant matches are affected by strategy, agent composition, map choices, daily form, and in-series adaptation. These factors create uncertainty that cannot be fully captured by historical statistics alone.

Main Findings

- Pre-match historical features provide a moderate predictive signal, with the best ROC-AUC around 0.66.
- Differential features were the most useful because they compare the two teams directly.
- Context features such as elimination status and grand finals were less influential than expected.
- Logistic Regression remained competitive, which suggests that the available feature space contains a fairly linear signal.
- XGBoost produced the best F1 score, but it is harder to explain than Logistic Regression.

Model Tradeoff

No single model is perfect. Logistic Regression is best when the stakeholder needs a clear explanation of why a prediction was made. XGBoost is better when the focus is raw classification performance. Random Forest provides a middle ground because it can model nonlinear patterns and still offer feature importance, but in this project it performed weaker than the other two main models on the 2025 test set.

Limitations, Sensitivity Analysis and Conclusion

Limitations

- The model uses historical pre-match features, so it does not fully capture agent composition quality, map veto strategy, round-by-round momentum, or player-level form.
- There may be temporal distribution shift between seasons because teams change rosters, strategies, and coaches over time.
- New teams have less historical data, which creates a cold-start problem.
- The model predicts match outcomes, not individual map outcomes, so it cannot fully represent in-series adaptation.

Sensitivity Analysis

Sensitivity analysis was used to test how model performance changed when important hyperparameters were adjusted. For Random Forest, `max_depth` was examined because deeper trees can overfit. For XGBoost, `learning_rate` was examined because aggressive learning can create a large train-test gap. The final settings were selected to balance predictive performance with generalization to future-season data.

Conclusion

This project built a machine learning pipeline for predicting Valorant Champions Tour match outcomes using three chronological Kaggle data subsets from 2023, 2024, and 2025. The task was correctly framed as binary classification, and three relevant machine learning models were compared: Logistic Regression, Random Forest, and XGBoost.

The results showed that historical team strength features, especially differential features, can predict match outcomes better than random guessing. Logistic Regression offered the best explanation and strongest ROC-AUC, while XGBoost produced the best F1 score. This supports the main aim of the B-Rank mission: comparing the strengths and weaknesses of different approaches rather than only choosing one model.

Future improvements could include map-specific modelling, player-level form, agent composition features, roster change indicators, and Elo-style opponent strength ratings. These additions could make the predictions more accurate and more useful to esports stakeholders.